

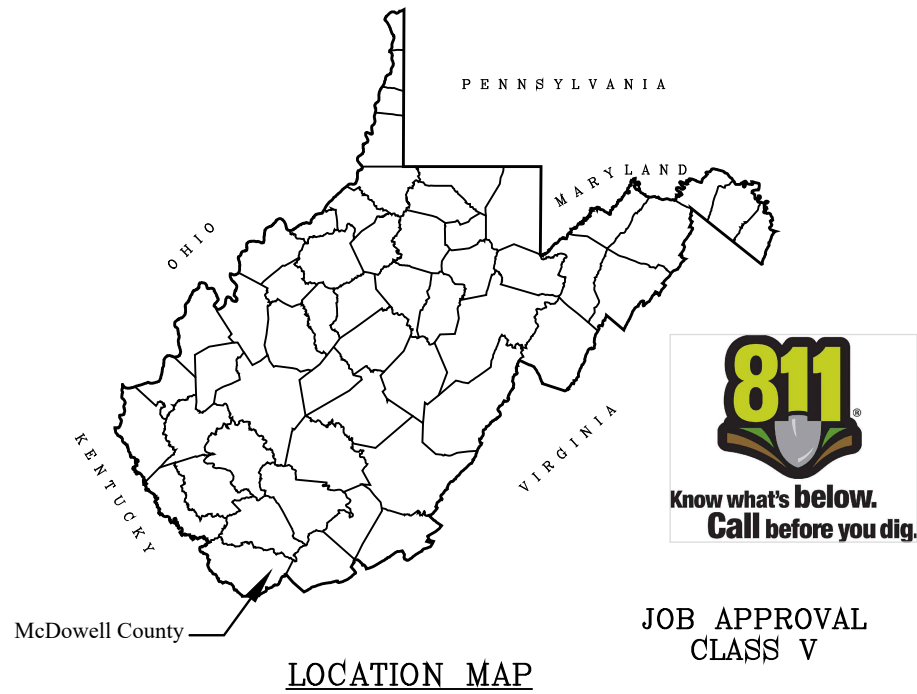
DRAWINGS PREPARED UNDER THE AUTHORITY OF THE
EMERGENCY WATERSHED PROTECTION ACT
by the
NATURAL RESOURCES CONSERVATION SERVICE
of the
UNITED STATES DEPARTMENT OF AGRICULTURE
STREAMBANK STABILIZATION
ELKHORN CREEK, SITES 1 & 2
DAMAGE SURVEY REPORT NO. 54-02-25-5087-001
MCDOWELL COUNTY, WEST VIRGINIA

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UTILITY NOTIFICATION

"The Natural Resources Conservation Service makes no representation as to the existence or non-existence of any utilities at the construction site. Shown on these construction drawings are those utilities which have been identified. It is the responsibility of the contractor(s) to assure themselves that no hazard exists or damage will occur to utilities."



<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%;">Designed <u>A. DEICHERT</u></td> <td style="width: 50%;">Date <u>5/2025</u></td> </tr> <tr> <td>Drawn <u>N. KNOX</u></td> <td>5/2025</td> </tr> <tr> <td>Checked <u>PM, DS, RS, TS</u></td> <td>6/2025</td> </tr> <tr> <td>Approved <u>A. DEICHERT</u></td> <td>7/2025</td> </tr> </table>	Designed <u>A. DEICHERT</u>	Date <u>5/2025</u>	Drawn <u>N. KNOX</u>	5/2025	Checked <u>PM, DS, RS, TS</u>	6/2025	Approved <u>A. DEICHERT</u>	7/2025	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%;">File Name</td> <td style="width: 50%;">Drawing Name</td> </tr> <tr> <td colspan="2" style="height: 40px;"></td> </tr> </table>	File Name	Drawing Name			2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK, SITES 1 & 2 COVER SHEET
Designed <u>A. DEICHERT</u>	Date <u>5/2025</u>													
Drawn <u>N. KNOX</u>	5/2025													
Checked <u>PM, DS, RS, TS</u>	6/2025													
Approved <u>A. DEICHERT</u>	7/2025													
File Name	Drawing Name													
United States Department of Agriculture Natural Resources Conservation Service														

CONSTRUCTION NOTES

1. Call 811 to locate utilities prior to mobilization and excavation.
2. Access roads shall be restored to pre-construction condition or better.
3. All fences removed and/or damaged as a result of the contractor's operations shall be restored to a condition equal to or better than they existed before the start of work.
4. Soil compacted as a result of construction operations shall be loosened for a depth of 6-inches below the compacted soil layer and re-graded to blend with existing ground.
5. Provide sub-channel as shown on Typical Channel Bottom Configuration drawing.
6. While working in the stream all efforts will be made to reduce downstream impacts including keeping the excavator tracks elevated above flowing water. A diversion berm shall be established at the top of each stream access point to minimize surface water from flowing toward the stream.
7. Required pollution control measures shall be installed as needed in accordance with Construction Specification 5, Pollution Control.
8. Site access shall be off the main road and shall be in accordance with all West Virginia Department of Highways rules and guidelines. The contractor shall provide signage and flagmen for traffic control as required. See Construction Specification 9, Traffic Control.
9. Excess material from the excavation & obstruction removal shall be wasted off-site in accordance with all local, state, & federal regulations and as approved by the NRCS representative in the field. Floodplains will not be allowed as waste areas under any circumstances. Waste areas shall be seeded according to Construction Specification 6, Seeding, Sprigging and Mulching.
10. The contractor shall provide approved fill as required to complete the work.
11. Trash, tires, & woody debris within the work limits shall be disposed of off-site in an approved landfill. The contractor shall be responsible for the landfill tipping fees.
12. All pipes encountered in the embankment shall be extended through the rock riprap and/or gabions with the same pipe material or better to the face of the slope or baskets.
13. Pipes 12" diameter and larger shall be encased in a minimum of 6" of concrete when passing through rock riprap and/or gabions. Concrete shall have a minimum compressive strength of 2500 psi.
14. The contractor shall maintain the cleanliness of public and private roads impacted by hauling operations during the duration of the work.
15. Starting location of baskets will be marked in the field.
16. Reshape/transition stream banks at the upstream and downstream ends of stabilization measures as directed by NRCS representative.

SITE 1 - TABLE OF ESTIMATED QUANTITIES

GEOTEXTILE SY*	BLANKET DRAIN TONS	DRAIN PIPE LF	DRAIN PIPE PERFORATED LF	DRAIN PIPE TEE EA	DRAIN PIPE CAP EA	RIPRAP TONS	GROUT RIPRAP CY	FOUNDATION PREP SHALLOW BEDROCK SY**	EARTH FILL CY
500	79	25	10	5	10	179	36	30	AS NEEDED

* THE AREA REQUIRED FOR OVERLAP IS NOT INCLUDED IN THE QUANTITY FOR GEOTEXTILE.

** QUANTITIES ASSUMING BEDROCK ENCOUNTERED FOR ENTIRE AREA.

SITE 2 - TABLE OF ESTIMATED QUANTITIES

GABION GEOTEXTILE SY*	RIPRAP TONS	GABIONS CY	EARTH FILL CY	FOUNDATION PREP SHALLOW BEDROCK SY**	FOUNDATION PREP ROCK RIPRAP TONS***	FOUNDATION PREP GEOTEXTILE SY***
155	0	160	AS NEEDED	62	186	248

* THE AREA REQUIRED FOR OVERLAP IS NOT INCLUDED IN THE QUANTITY FOR GEOTEXTILE.

** QUANTITIES ASSUMING BEDROCK ENCOUNTERED FOR ENTIRE AREA.

*** QUANTITIES ASSUMING ROCK RIPRAP FOUNDATION FOR ENTIRE AREA.

File Name	Date	Designed	Drawn	Checked	Approved
	5/2025	N. KNOX			
2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK, SITES 1 & 2 QUANTITIES AND CONSTRUCTION NOTES					
United States Department of Agriculture  Natural Resources Conservation Service					

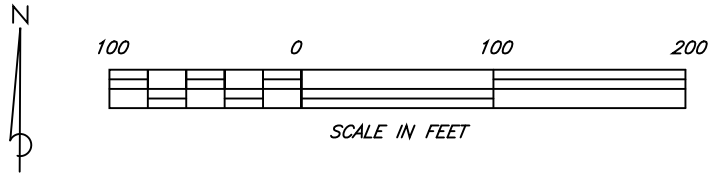


2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK, SITES 1 & 2 LOCATION MAP		Designed Drawn N. KNOX Checked Approved	Date 5/2025	File Name Drawing Name
United States Department of Agriculture USDA		Natural Resources Conservation Service		
		Sheet 3 of 14		

PROJECT CONTROL

DESCRIPTION	NORTHING	EASTING	ELEVATION
EXISTING TBM	151717.77	1844064.98	1721.25
- DUPLEX NAIL IN POWER POLE			

NAD83 WV STATE PLANE, SOUTH ZONE, US FOOT GRID. NAVD88 VERTICAL DATUM



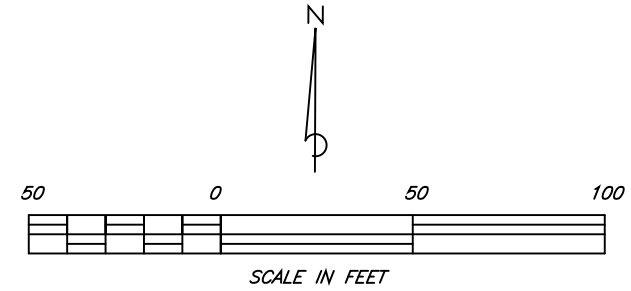
File Name		Date	
Drawing Name		5/2025	
Designed	N. KNOX	Drawn	
Checked		Approved	
2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK, SITES 1 & 2 SITE 1 - SITE MAP			
United States Department of Agriculture 		Natural Resources Conservation Service	

LEGEND

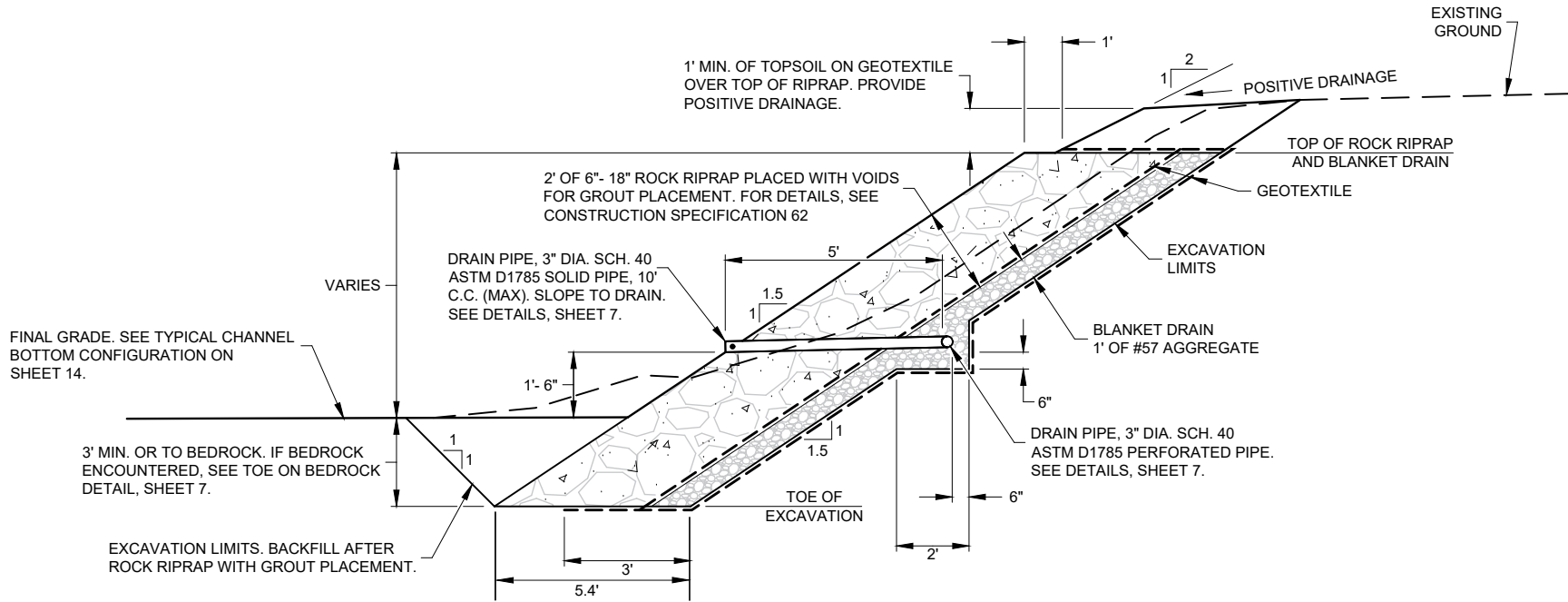
	EDGE OF PAVED PARKING
	TOP OF BANK
	TOE OF SLOPE
	CL CONSTRUCTION LIMITS
	EDGE OF ACCESS ROAD
	CENTERLINE OF STREAM
	CONTROL POINT

NOTES

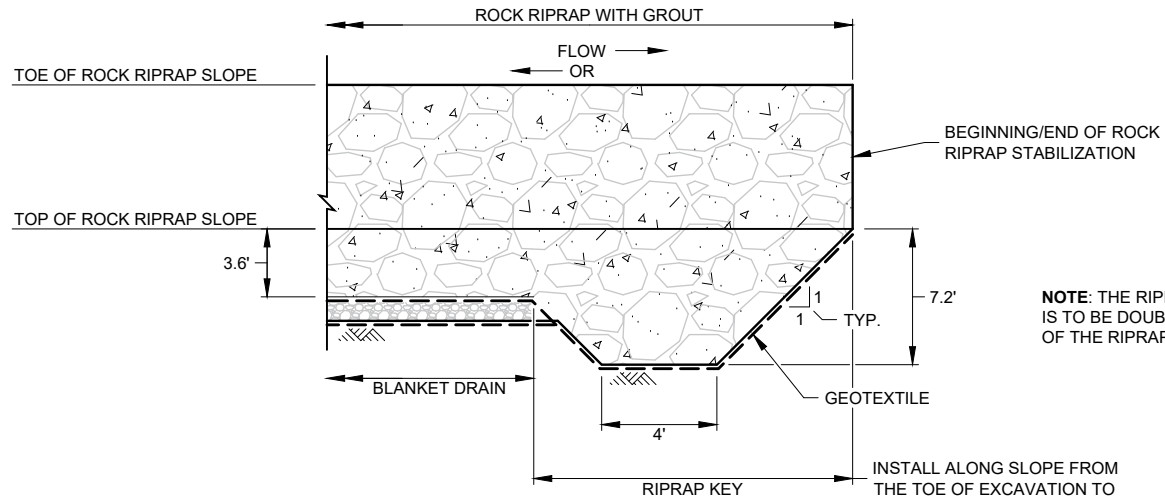
1. CHANNEL CLEARING AND DEBRIS DISPOSAL SHALL BE DIRECTED BY NRCS REPRESENTATIVE.
2. BORROW MATERIAL SHALL BE TAKEN FROM WITHIN THE CONSTRUCTION LIMITS DIRECTED BY THE NRCS REPRESENTATIVE.
3. DISPOSAL AREA FOR EXCAVATED MATERIAL SHALL BE AT AN APPROVED LOCATION OFFSITE.



 United States Department of Agriculture Natural Resources Conservation Service	File Name	Drawing Name	Sheet 5 of 14
	Date	5/2025	
2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK, SITES 1 & 2 SITE 1 - PLAN VIEW			
Designed N. KNOX Drawn N. KNOX Checked Approved			



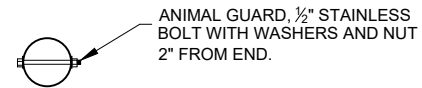
SECTION - GROUTED ROCK RIPRAP



PLAN - RIPRAP KEY

NOT TO SCALE

File Name	Date	Designed	Drawn	Checked	Approved
	5/2025	N. KNOX			
Drawing Name	2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK SITES 1 & 2 TYPICAL GROUTED ROCK RIPRAP SECTION & DETAILS				
Sheet 6 of 14	United States Department of Agriculture USDA Natural Resources Conservation Service				



PERFORATED PIPE



NOT TO SCALE

 United States Department of Agriculture	2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK SITES 1 & 2 TYPICAL GROUTED ROCK RIPRAP DETAILS	Designed _____ Drawn N. KNOX 5/2025	Date _____	File Name _____
		Checked _____ Approved _____	Drawing Name _____	Sheet 7 of 14

PROJECT CONTROL

DESCRIPTION	NORTHING	EASTING	ELEVATION
EXISTING TBM - PK NAIL	140812.73	1847498.43	1878.23

NAD83 WV STATE PLANE, SOUTH ZONE, US FOOT GRID. NAVD88 VERTICAL DATUM



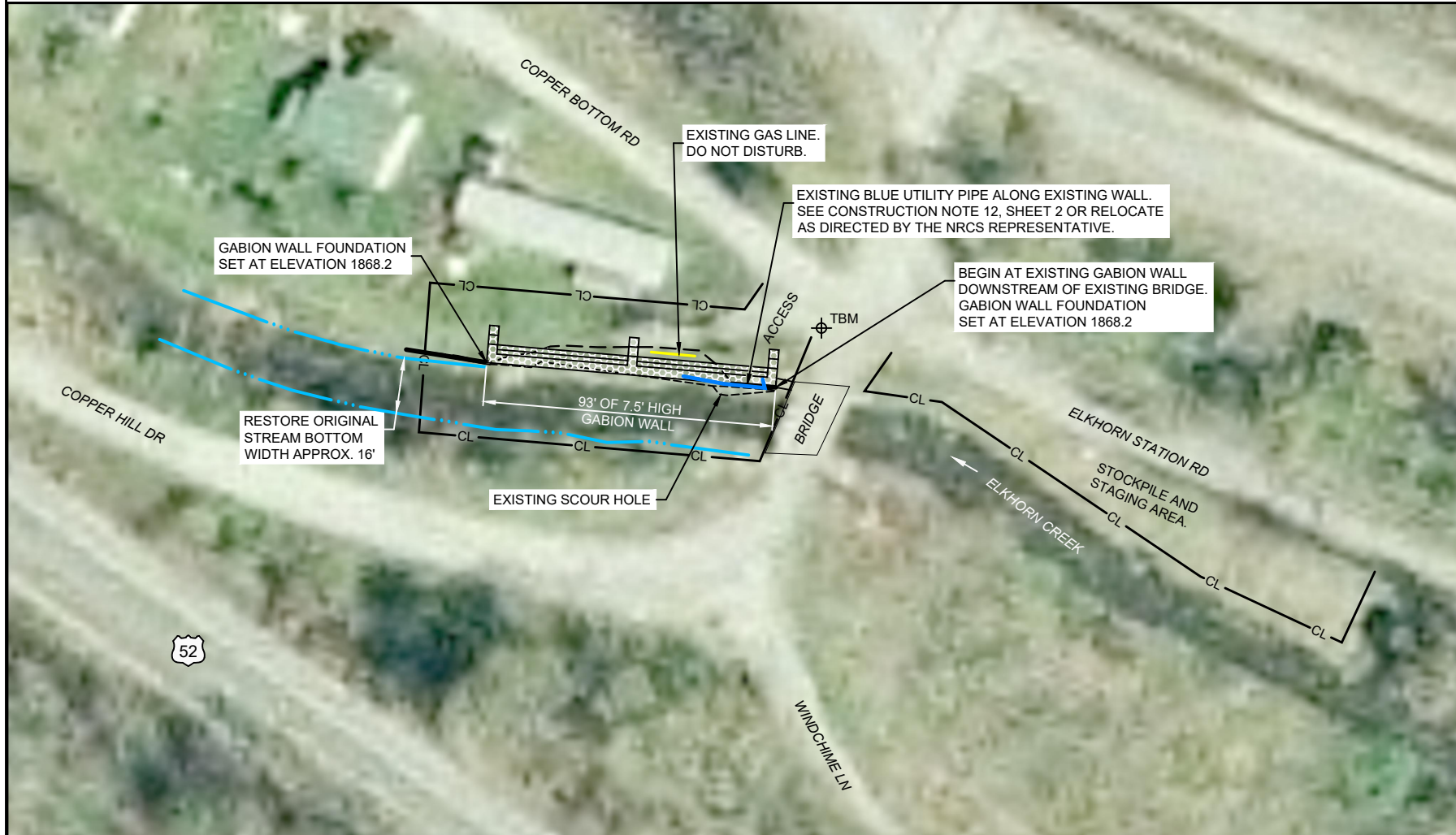
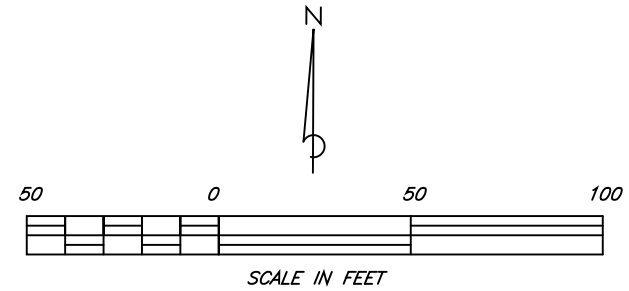
File Name		Date	
Designed		5/2025	
Drawn N. KNOX		5/2025	
Checked			
Approved			
2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK, SITES 1 & 2 SITE 2 - SITE MAP			
United States Department of Agriculture 		Natural Resources Conservation Service	

LEGEND

	EXISTING WALL
	TOP OF BANK
	TOE OF SLOPE
	CL
	CONSTRUCTION LIMITS
	EDGE OF STREAM
	CONTROL POINT

NOTES

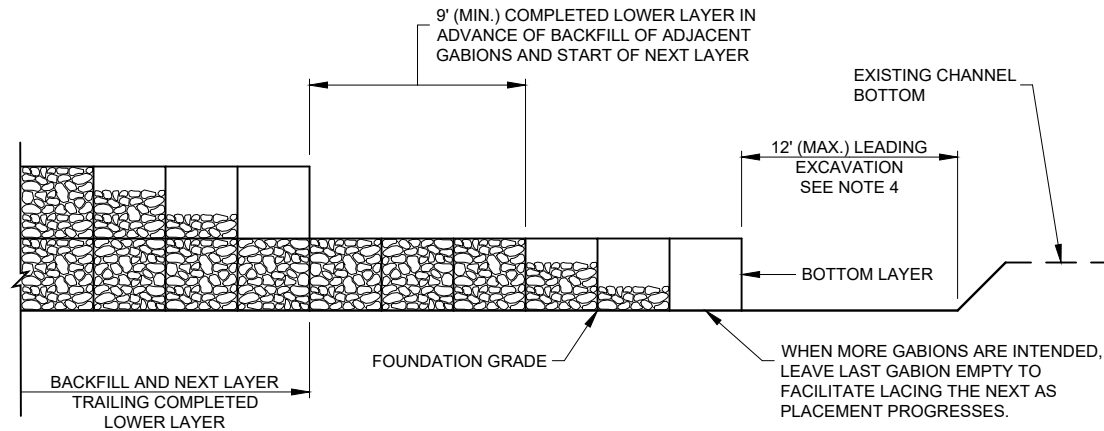
1. EXCAVATED MATERIAL NOT SUITABLE FOR BACKFILL, DEBRIS AND REQUIRED WALL REMOVAL WITHIN THE CONSTRUCTION LIMITS SHALL BE DISPOSED OF AT AN APPROVED LOCATION OFFSITE.
2. GABION WALL ALIGNMENT AND LAYOUT LOCATION MAY VARY AS DETERMINED BY THE NRCS REPRESENTATIVE.
3. BACKFILL MATERIAL BEHIND GABIONS SHALL BE FROM CHANNEL SHAPING AND/OR FROM EXCAVATION AS DETERMINED BY THE NRCS REPRESENTATIVE. BACKFILL MATERIAL SHALL BE FREE OF TRASH, CONCRETE, WOODY DEBRIS, ETC.
4. CONTRACTOR SHALL CONSERVE SUITABLE MATERIALS FOR FINAL GRADING AND SEEDING OPERATIONS.



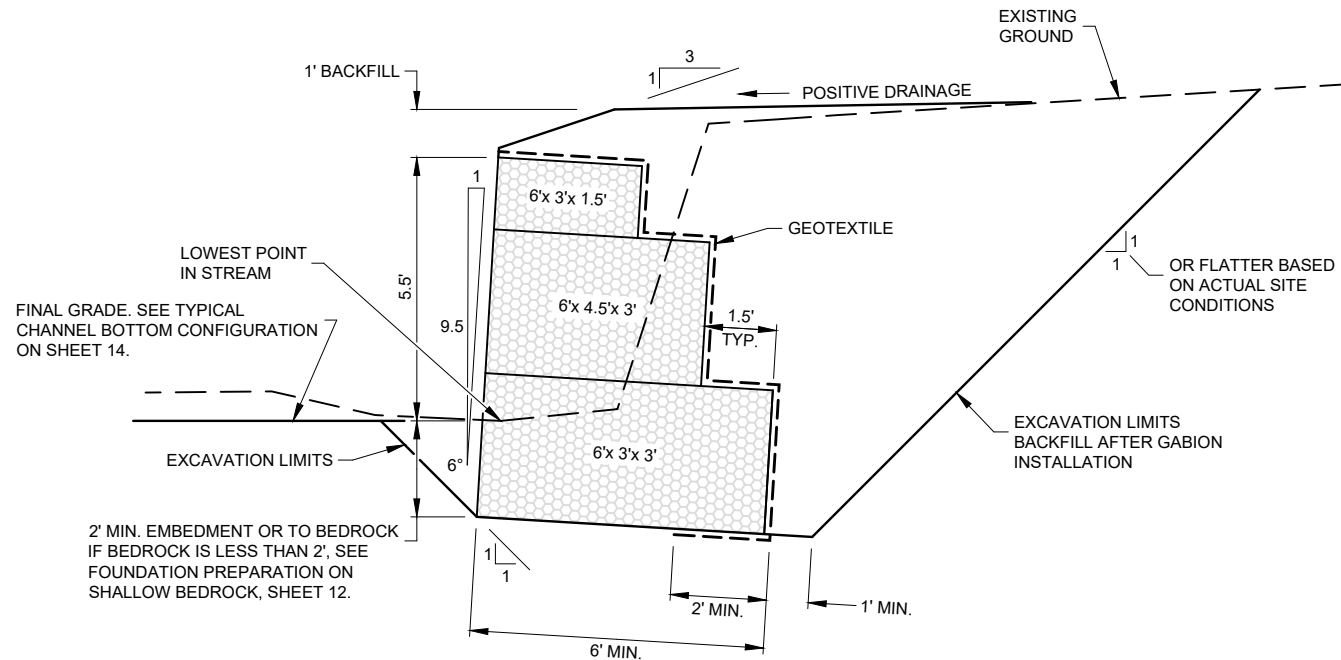
	United States Department of Agriculture	2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK, SITES 1 & 2 SITE 2 - PLAN VIEW	File Name Date Designed Drawn N. KNOX Checked Approved	Drawing Name 5/2025 Sheet 9 of 14
	Natural Resources Conservation Service			

NOTES

1. SEE TYPICAL CHANNEL BOTTOM CONFIGURATION SHEET 14.
2. REFER TO SHEET 2 FOR ESTIMATED QUANTITIES.
3. AFTER THE FOUNDATION IS EXCAVATED, NRCS REPRESENTATIVE SHALL DETERMINE IF FOUNDATION TREATMENT WILL BE REQUIRED AND TYPE. SEE CONDITIONAL FOUNDATION DETAILS ON SHEETS 12 & 13.
4. THE CONTRACTOR SHALL ONLY EXCAVATE THE AMOUNT OF FOUNDATION NEEDED TO INSTALL THE BOTTOM LAYER OF GABION WALL INTENDED TO PLACE PER MANUFACTURER'S RECOMMENDATIONS DURING THE SAME WORKDAY.
5. BACKFILL BEHIND EACH LAYER OF GABIONS AS THE LAYERS PROGRESS VERTICALLY.



**PROFILE - GABION
PLACEMENT/FILLING SEQUENCE**



SECTION - 7.5' HIGH GABION WALL

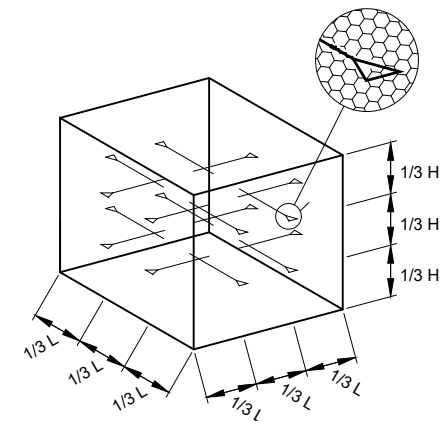
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File Name	Date	Designed	Drawn	Checked	Approved
	5/2025	N. KNOX	N. KNOX		
2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK, SITES 1 & 2 TYPICAL GABION WALL SECTION & DETAILS					
United States Department of Agriculture			Natural Resources Conservation Service		



-
- Diagram illustrating the repair of a wire mesh. A wire is shown doubled back and twisted, forming a continuous loop across the mesh. The wire is labeled "WIRE DOUBLED BACK AND TWISTED". The diagram shows the wire passing through the mesh, with dimensions indicating a 5" section on the left and a 5" section on the right.

LACING DETAIL

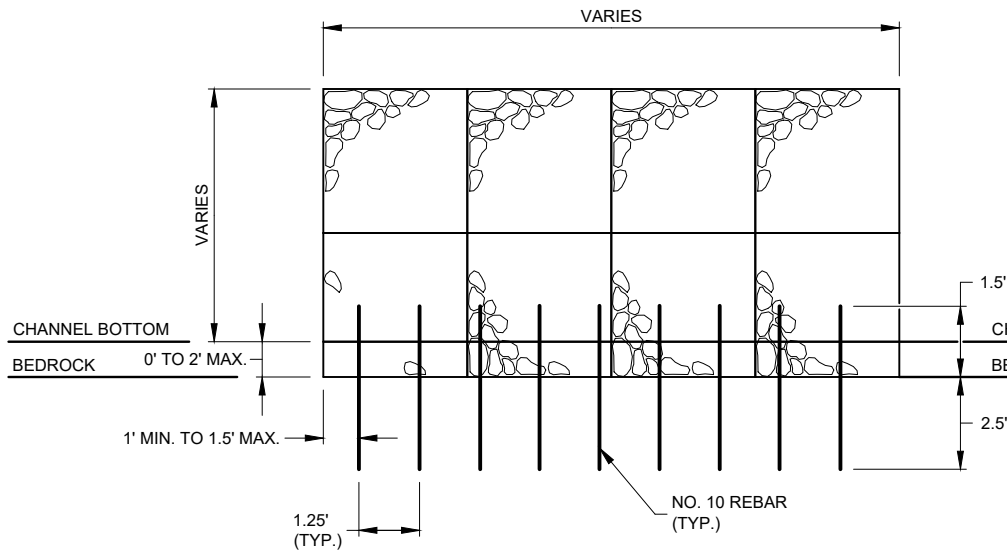


INTERIOR SUPPORT DETAIL

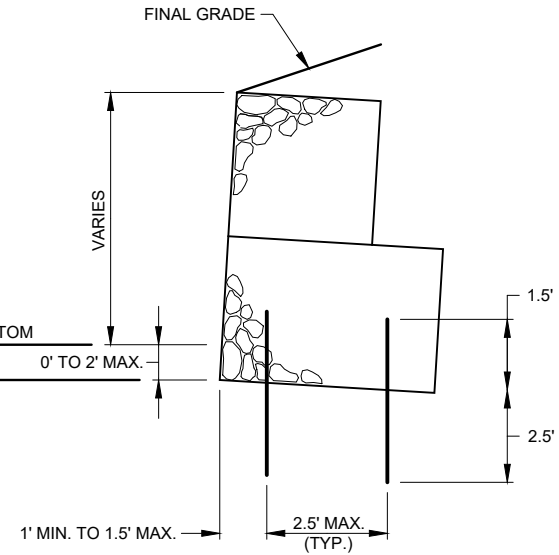
 United States Department of Agriculture	2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK, SITES 1 & 2 TYPICAL GABION WALL DETAILS	Designed _____	Date _____	File Name _____
		Drawn <u>N. KNOX</u>	<u>5/2025</u>	Drawing Name _____
		Checked _____	_____	_____
		Approved _____	_____	Sheet 11 of 14

NOTES:

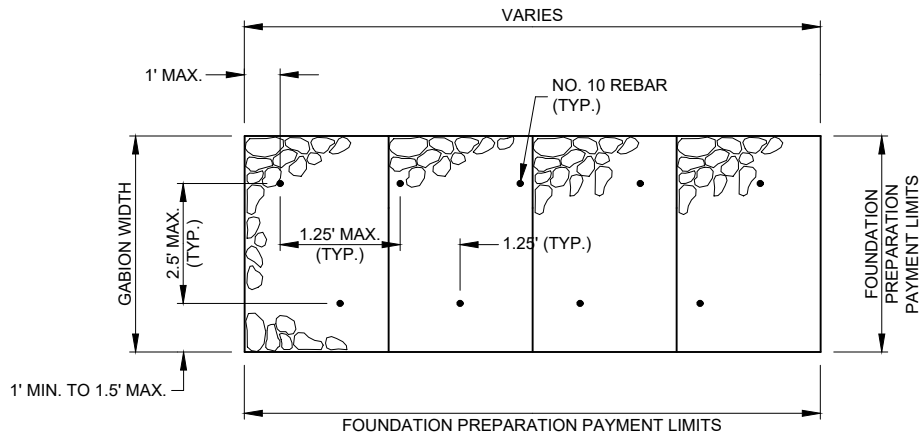
1. Drill holes for Rebar shall be 1 3/8" diameter. The use of epoxy or non-shrink grout ***is not*** required. The rebar shall be driven by hammer to the required depth without bending.
2. Care shall be taken by the contractor to prevent damage to the rebar during gabion filling operations.
3. Different gabion bottom layer configurations will require additional rebar by repeating the pattern shown. Quantities on sheet 2 based on bottom width x length.



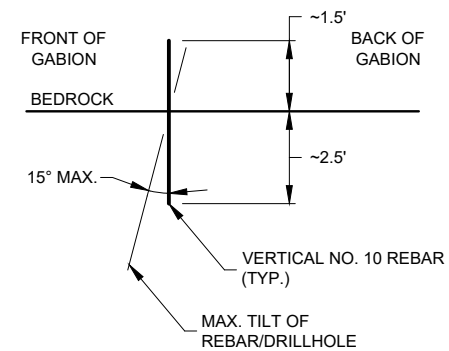
**PROFILE - GABIONS
WITH REBAR SPACING**



**SECTION - GABIONS
WITH REBAR SPACING**



**PLAN - GABIONS WITH
REBAR/DRILLHOLE PATTERN**

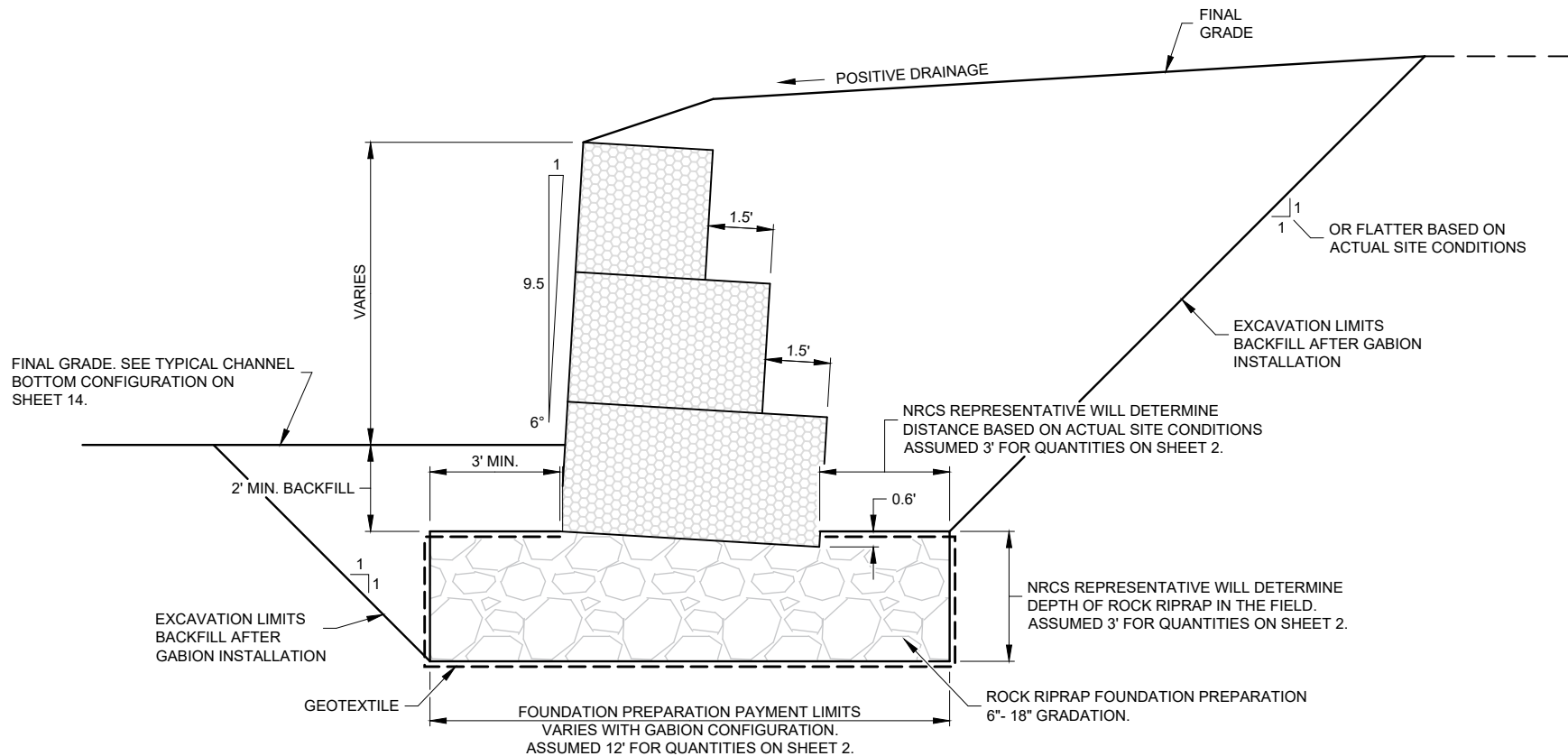


**MAXIMUM ANGLE OF
REBAR/DRILLHOLE PATTERN**

NOTE: Final pattern and spacing to be directed by NRCS representative in the field.

NOT TO SCALE

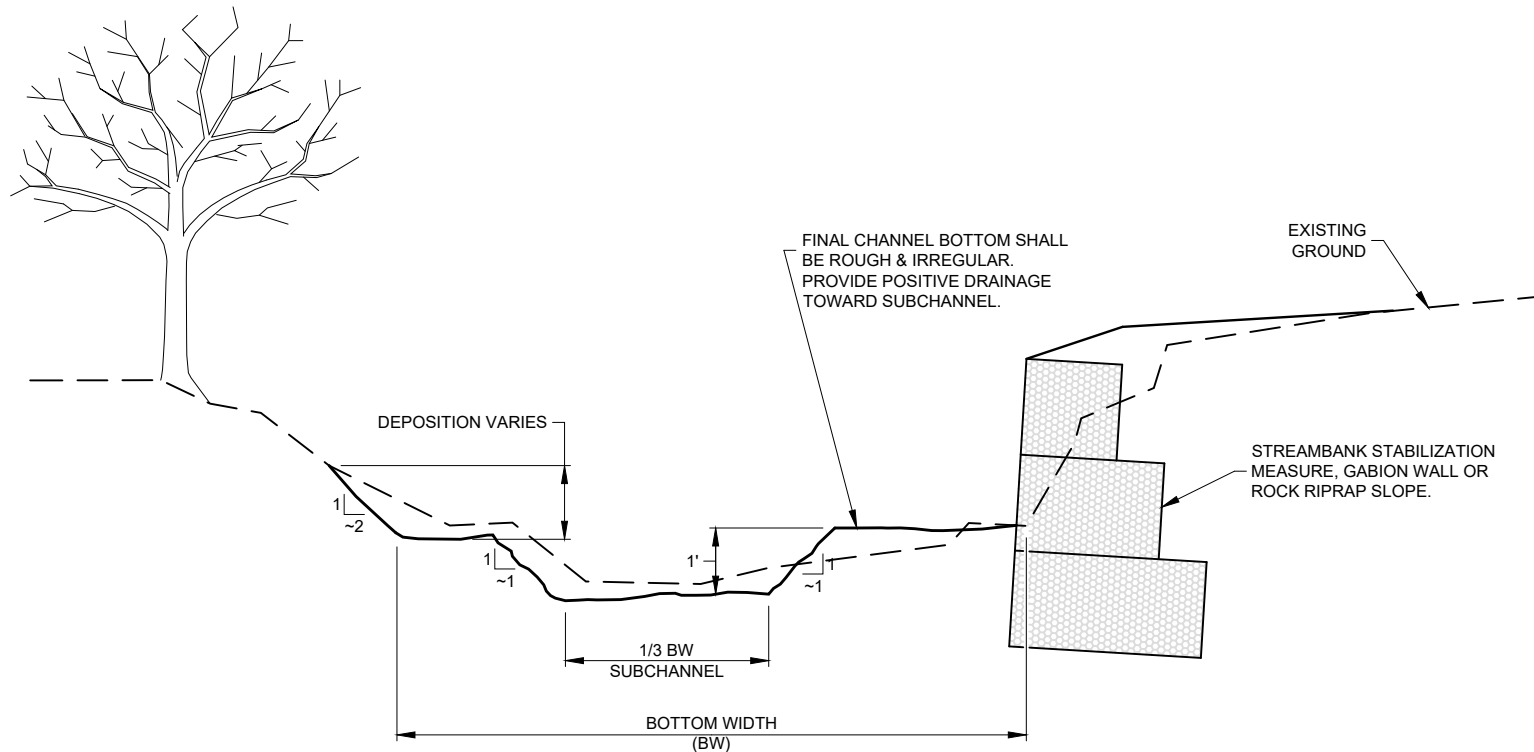
File Name	Date	Designed	Drawn	Checked	Approved
		N. KNOX	5/2025		
2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK, SITES 1 & 2 TYPICAL FOUNDATION PREPARATION ON SHALLOW BEDROCK DETAIL					
United States Department of Agriculture			Natural Resources Conservation Service		



SECTION - ON RIPRAP FOUNDATION

NOT TO SCALE

 United States Department of Agriculture Natural Resources Conservation Service	2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK, SITES 1 & 2 TYPICAL SECTION ON RIPRAP FOUNDATION	
Designed _____ Drawn N. KNOX Checked _____ Approved _____	Date _____ 5/2025	File Name _____ Drawing Name _____ Sheet 13 of 14



CHANNEL BOTTOM CONFIGURATION
SEE SITE PLAN

Notes:

1. Final location of subchannel to be located in the field by the NRCS representative. The subchannel shall be transitioned as necessary to meet existing grades at the upstream and downstream site extents.
2. Subchannels are not required for bedrock streambeds.
3. Use material from the required excavations and/or approved fill from offsite to reconstruct streambanks as necessary. Seed and mulch per Construction Specification 6, Seeding, Sprigging and Mulching.

NOT TO SCALE

 United States Department of Agriculture	Natural Resources Conservation Service	
	2025 EWP STREAMBANK STABILIZATION MCDOWELL COUNTY, WEST VIRGINIA ELKHORN CREEK, SITES 1 & 2 TYPICAL CHANNEL BOTTOM CONFIGURATION	
Designed _____ Drawn N. KNOX Checked _____ Approved _____	Date _____ 5/2025	File Name _____ Drawing Name _____ Sheet 14 of 14